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Mapping Paddy Lands using Remote Sensing Techniques and Land Use Land Cover Change Detection Over a Decade in Gampaha District, Sri Lanka

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Abstract

Rice production constitutes a critical component of food security across numerous Asian nations. However, rapid urbanization has significantly constrained the availability of agricultural land resources. This study evaluated the efficacy of remote sensing (RS) techniques for generating comprehensive paddy land maps across three distinct phenological stages—initial establishment, development, and harvesting—while simultaneously detecting land use land cover (LULC) transformations in Gampaha district, Sri Lanka, during the periods 2008/2009 and 2018/2019. Three remote sensing methodologies were systematically assessed: (1) supervised classification, (2) Tasselled Cap Transformation integrated with Normalized Difference Vegetation Index (NDVI), and (3) Land Surface Water Index (LSWI) combined with NDVI analysis. LULC change detection was conducted utilizing Landsat 5 and Landsat 8 satellite imagery processed through ArcGIS 10.8 software. Classification accuracy assessments revealed overall accuracies of 85.5% and 89.5% for supervised classification, 84.0% and 88.5% for Tasselled Cap-NDVI methodology, and 85.5% and 53.0% for LSWI-NDVI approach across the 2008/2009 and 2018/2019 periods, respectively. Results demonstrated that supervised classification proved most effective for identifying paddy fields during the harvesting stage, while Tasselled Cap-NDVI methodology exhibited superior performance during the development stage, and LSWI-NDVI analysis showed optimal results for initial stage detection. Temporal analysis revealed substantial LULC transformations during the study period, with built-up areas experiencing a 126.4% increase, concurrent with a dramatic 76.5% reduction in paddy cultivation areas. These findings underscore the severe pressure of urban expansion on agricultural land resources in the region.

Introduction

Agriculture is the most important sector of a country, playing a pivotal role in improving and guaranteeing the availability of food, which is also known as food security (Pawlak & Kołodziejczak, 2020). Food security is one of the primary goals of all countries, aiming to eliminate hunger, alleviate undernutrition, reduce poverty, and promote economic development (Bandumula, 2018).

Rice is one of the primary agricultural food crops consumed by nearly half of the world's population, and a significant percentage of the global population (Bandumula, 2018; Mohidem et al., 2022). Asia accounts for approximately 90% of global rice cultivation and consumption, with China and India being the world's largest producers (Bandumula, 2018).

In recent years, overall rice consumption has increased slightly worldwide. For instance, in 2008/2009, 437.18 million metric tons of rice were consumed, which then reached approximately 485 million metric tons within ten years (Shahbandeh, 2024). This development could be due to the increase in the human population. To meet the demand for rice, rice production needs to increase worldwide.

Paddy cultivation, which occupies 34% of the total cultivated area in Sri Lanka (Malkanthi et al., 2019; Rice Research & Development Institute, 2022), is a major agricultural sector that has been practiced

since ancient times. Rice plays a crucial role in the country's economy and the lives of its people as a primary staple food (Sugirtharan & Delina, 2013).

There are two seasons, Yala and Maha, which are associated with two main monsoons, during paddy field cultivation in Sri Lanka. The main season is Maha, September to March when the northeast monsoon is activated. The Yala season is associated with the southwest monsoon period, which spans from May to August (Aponsu, 2021). Since independence was achieved in 1948, agricultural policies in Sri Lanka have focused primarily on improving the self-sufficiency of all food crops. In particular, the target for producing rice was set at 100% of the domestic demand (Davis et al., 2016). However, increased human demand and climate variability have created greater concern about global food trading and its impacts on nations' food security. Sri Lanka is ranked 66th in terms of the global food security index.

Increasing crop intensity and national average yields are the main options available for achieving good food security and satisfying the high food demand and production targets. Therefore, identifying available paddy fields and their distribution throughout the country is highly important for ensuring the actual production of rice and increasing rice production during the initial stages.

Satellite remote sensing is the most common source of data for the identification, quantification, and mapping of Land Use Land Cover (LULC) patterns and changes (Wijesundara et al., 2023) due to its repetitive data collection, computer-friendly digital format, and precise geo-referencing procedures (Abd El-Kawy et al., 2011).

As rice is one of the world’s major staple foods, mapping the extent of paddy land using satellite images in many countries is important. Paddy land mapping using different remote sensing techniques has been performed on a subcontinental scale, including in Southern and Southeastern Asia (Gumma, 2011; Xiao et al., 2006), and on a local scale, MODIS-based paddy land maps in countries such as India, China, and Bangladesh (Gumma, 2011; Sun et al., 2009; Xiao et al., 2005).

Compared to the related literature, research on identifying and mapping paddy lands in Sri Lanka using satellite images is scarce. A recent study, limited to a specific region of Sri Lanka, assessed agricultural land changes in the Jaffna Peninsula using Landsat TM and Landsat OLI imagery from 1988, 2004, 2009, 2013, and 2018 (Gopalakrishnan et al., 2020).

Therefore, the overall aims of this study were (1) to identify the most appropriate and accurate remote sensing technique for the identifying of each paddy growth stage: initial, development, and harvesting stage, and (2) to assess how land use land cover patterns have transformed over ten years, from 2008/2009 to 2018/2019, in one of the highly urbanized areas in Sri Lanka, which has been subjected to a reduction in agricultural lands mainly paddy lands with industrialization and economic growth in recent years.

Methodology

Study area

The study was conducted in the Gampaha district, which is one of the most populated districts in Sri Lanka (Fig. 1). It is a low-country wet zone district located in the west of the country bounded by the Kelani River to the South, Ma Oya to the North, the Kegalle district to the East, and the Indian Ocean to the West. The mean annual rainfall is almost 2500 mm, and the area is characterized by built-up areas, plantations, wetlands, agricultural lands, and bare lands.

Data collection

Shapefiles of administrative boundaries in Sri Lanka were obtained from the Department of Survey and Land Registration, Colombo.

For the time series analysis, Landsat 8 OLI/TIRS Level 2 images (2018, 2019) and Landsat 5 TM Level 2 images (2008, 2009) were ordered and downloaded from the USGS Earth Explorer. (http://earthexplorer.usgs.gov/). Three images from three growth stages of paddy field were taken during the Maha season for both years. The acquisition dates of the satellite images for the years 2008/2009 were 04/11/2008, 23/01/2009, and 08/02/2009. For the years 2018/2019, they were 02/12/2018, 03/01/2019, and 20/02/2019.

Paddy Land Identification

Three methods were used to find the most suitable remote sensing techniques for identifying and mapping paddy lands according to their growth stages namely the initial, development, and harvesting stages, using Landsat images analyzed in ArcGIS 10.2. The methods were (1) Supervised classification method, (2) Tasseled Cap Transformation/ TCT and Normalized Difference Vegetation Index/ NDVI method, and (3) Land Surface Water Index/ LSWI and NDVI method.

1. Supervised classification method

Radiometrically calibrated, orthorectified, and high-quality Level 2 Landsat satellite scenes for the years 2008, 2009, 2018, and 2019 were obtained and downloaded from the USGS Earth Explorer (http://earthexplorer.usgs.gov/). The steps taken for Method 1 are shown in Fig. 2.

A useful single-band raster dataset was selected and combined to create a multiband raster dataset. Following the composition, the study area was extracted using the Gampaha district shape file.

The composite image was classified into four classes: water bodies, built-up areas, paddy lands, forests, and other vegetation using the supervised classification (Maximum likelihood classification) method. After that, the obtained classified images were reclassified to create a Land Use Land Cover (LULC) map with only two classes: paddy and nonpaddy.

All the images downloaded in 2008/2009 and 2018/2019 were subjected to Supervised classification, and the spectral signatures obtained from the training samples were used to classify the images. Google

Earth Engine was useful in identifying paddy lands because they have a unique geometric shape, usually rectangular. Man-made structures, such as roads and canals, can be identified as straight lines using Google Earth Engine.

The spectral variations observed in the paddies were analyzed using the Google Earth Engine timeline, and subsequently utilized to identify distinct growth stages. First, the paddy lands appear as flooded areas, converted to light green areas during development, and yellow at the harvesting stage. Evergreen forestlands are dark green, built-up areas are gray due to the extensive use of concrete, and water usually appears black due to light absorption.

Among the three maps created in a single year, the best map from each year was selected based on an accuracy assessment.

2. Tasseled Cap Transformation (TCT) and the Normalized Difference Vegetation Index (NDVI) method

Figure 3 illustrates the steps taken to classify the Landsat images from 2008/2009 and 2018/2019 using Method 2.

Tasseled Cap Transformation (TCT) is a mathematical method used in remote sensing to identify agricultural and ecological characteristics. Three uncorrelated indices: brightness (BRT), greenness (GRN), and wetness (WET) are produced by converting covariant data in satellite images (Sheng et al., 2011).

The BRT gives a gray tone gradation of non-vegetation to water. The gradation of vegetation cover, from dense to rare, is indicated by the GRN, while areas of water presence are indicated by the WET. This method can be used to identify the paddies that are in the growing stage (Riadi & Suriadi, 2017).

For the mathematical calculation of TCT, the reflectance and coefficient values of each band of the Landsat image were used. According to Riadi & Suriadi, (2017), the following mathematical equations (Eqs. 1–3) were used to obtain the BRT, GRN, and WET from Landsat 8 images.

$$Brightness = B2 * .3029 + B3 * .2786 + B4 * .4733 + B5 * .5599 + B6 * .508 + B7 * .1872 \quad (1)$$

$$Greenness = B2 * .2941 + B3 * -.243 + B4 * -.5424 + B5 * .7276 + B6 * .0713 + B7 * -.1608 \quad (2)$$

$$Wetness = B2 * .1511 + B3 * .1973 + B4 * .3283 + B5 * .3407 + B6 * -.7117 + B7 * -.4559 \quad (3)$$

According to Liu et al. (2016), the following mathematical equations (Eqs. 4–6) were used to derive BRT, GRN, and WET from Landsat 5 images.

$$Brightness = B1 * .2043 + B2 * .4158 + B3 * .5524 + B4 * .5741 + B5 * .3124 + B7 * .2303 \quad (4)$$

$$Greenness = B1 * -.1603 + B2 * -.282 + B3 * -.4934 + B4 * .794 + B5 * -.0002 + B7 * -.145 \quad (5)$$

$$Wetness = B1 * .0315 + B2 * .2021 + B3 * .3102 + B4 * .1594 + B5 * -.6806 + B7 * -.6109 \quad (6)$$

(where B1 = Band 1, B2 = Band 2, B3 = Band 3, B4 = Band 4, B5 = Band 5, B6 = Band 6, B7 = Band 7)

Normalized Difference Vegetation Index (NDVI) is a widely used quantitative metric for indicating vegetation cover and crop status (Gandhi et al., 2015). It is calculated as the difference in the ratio between the measured reflectances in red and near-infrared bands. It supports monitoring vegetation, drought assessment, and evaluating crop cover and agricultural conditions at both global and local levels (Bhandari et al., 2012).

The following Eq. 7 and Eq. 8 equations were used to obtain the NDVI.

$$NDVI = Band4 - Band3/Band4 + Band3 \quad (7)$$

(where Band 3 is the red band and Band 4 is the near-infrared band for Landsat 5)

$$NDVI = Band5 - Band4/Band5 + Band4 \quad (8)$$

(where Band 4 is the red band and Band 5 is the near-infrared band for Landsat 8)

By applying the above mathematical equations, Tasseled Cap Brightness (TCB), Tasseled Cap Wetness (TCW), and NDVI bands were created. The composite image was obtained by combining the three bands TCB, TCW, and NDVI.

The composite image was used to classify the image through supervised classification into four classes water bodies, built-up areas, paddies, and forests and other vegetation. A green color was given for paddies, a red to yellow color was given for forests, and other vegetation. A luminous green color was given for water bodies, and the purple color was given for built-up areas.

The classified images were reclassified to obtain a LULC map with only two classes: paddy and non-paddy. This method was applied to all the images, and accuracy was assessed using Google Earth data as the ground truth data. The best map was selected in each year 2008/2009 and 2018/2019 (Maha season).

3. Land Surface Water Index (LSWI) and NDVI

The steps taken to classify the Landsat images from 2008/2009 and 2018/2019 are shown in Fig. 4.

The Land Surface Water Index (LSWI) is a water index that uses the shortwave infrared (SWIR) and near-infrared (NIR) regions of the electromagnetic spectrum. The SWIR has a strong absorption of light by liquid water, and the LSWI is sensitive to the total amount of liquid water in the vegetation and its soil environment (Chandrasekar et al., 2010).

For the derivation of LSWI for Landsat 5 and 8 images, equations 9 and 10 were used, respectively.

$$LSWI = Band4 - Band7 / Band4 + Band7 \quad (9)$$

(where Band 4 is the near-infrared band and Band 7 is the shortwave infrared band for Landsat 5)

$$LSWI = Band5 - Band7 / Band5 + Band7 \quad (10)$$

(where Band 5 is the near-infrared band and Band 7 is the shortwave infrared band for Landsat 8)

In this study, a simple assumption was made for the detection of permanent water bodies. Pixels with NDVI values less than 0.1 were considered permanent water bodies and were masked, because they may interfere with paddy identification during flooding and the initial transplanting stage of paddy cultivation.

LSWI characteristics were investigated in relation to conventional NDVI characteristics, particularly during the early crop season (flooding and transplanting/initial).

The LSWI and NDVI were used to identify areas with high soil moisture and low vegetation coverage during the initial stage of paddy cultivation, where the LSWI was high and the NDVI was low. The condition LSWI > NDVI was used to identify paddy fields based on characteristics during the initial stage, where a low NDVI indicates paddy fields and a high NDVI indicates other vegetation (Sun et al., 2009; Xiao et al., 2005, 2006). Finally, a map representing paddy and non-paddy areas was obtained.

Accuracy Assessment

A significant step in the processing of remote sensing data is accuracy assessment or validation. It sets the information value of a user's resulting data.

The accuracy of the derived maps in 2008/2009 and 2018/2019 was assessed by using Google Earth reference maps as ground truth data. Two hundred ground truth points were constructed, with 100 from each class, paddy and non-paddy. Error-based matrices were developed to highlight the classification accuracy for the images in each year and for each method, which shows the relationship between the reference data and the derived map.

The kappa coefficients (K_{hat}) for the classified images were calculated using the following equation, Eq. 11, to determine the accuracy of the classification.

$$\mathrm{K_{hat}} = \frac{(\mathrm{Total} \times \mathrm{sum\ of\ correct}) - \mathrm{sum\ of\ (Row\ total} \times \mathrm{column\ total)}}{\mathrm{Total}^2 - \mathrm{Sum\ of\ (Row\ total} \times \mathrm{Column\ total)}}$$

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Land use land cover (LULC) change detection

For land use land cover (LULC) change detection from 2009 to 2019, 08/02/2009 and 20/02/2019 dates acquired Landsat 8 OLI/TIRS Level 2 and Landsat 5 TM Level 2 (2008,2009) satellite images were downloaded from the USGS Earth Explorer. (http://earthexplorer.usgs.gov/) respectively, considering the harvesting stage of paddy cultivation during the Maha season.

Then, multiband composite images for both 2009 and 2019, were created and corrected radiometrically to remove atmospheric noise, unnecessary data from the sensors (Withanage et al., 2024), and to increase the brightness magnitude (Seyam et al., 2023) using ArcGIS 10.8 software.

Supervised classification

To identify the land use land cover (LULC) for the 2009 and 2019 composite images, the supervised classification method was employed using the Maximum Likelihood Classification algorithm. The analysis was conducted in three steps: taking signatures or training sample selection from each LULC in the study area, classification, and accuracy assessment. The four classes were considered to map the study area as water bodies, built-up areas, forests and other vegetation, and paddy lands. More than 100 training samples were created for each LULC class. Then, the Maximum Likelihood classification, a widely used classifier for LULC mapping (Seyam et al., 2023) was subsequently employed following the accuracy assessment. To measure the accuracy, 400 randomly marked ground truth data points were gathered for each year using Google Earth.

Change detection

To calculate the change in each land use land cover area from 2009 to 2019, change detection analysis was performed in ArcGIS version 10.8. There, classified raster images were converted into vector layers, and the LULC area change for each class was calculated separately.

To calculate the percentage change (%) in each class, the following equation was used (Seyam et al., 2023)

$$\mathrm{Pi}\ (\%) = \frac{(\mathrm{Li} - \mathrm{Bi})}{\mathrm{Bi}} \times 100$$

P_i (%) indicates the percentage change in class i. B_i is the area of class i in the basic image (2009), and L_i is the area of class i in the latest image (2019).

Results

Derivation of paddy and non-paddy areas

Figure 5 shows the derived maps of paddy and non-paddy areas through three remote sensing techniques: (1) supervised classification method, (2) Tasseled Cap Transformation/ TCT and Normalized Difference Vegetation Index/ NDVI method, and (3) Land Surface Water Index/ LSWI and NDVI method for the years 2008/2009 and 2018/2019 during the Maha season. According to the maps, the area of paddy fields has decreased, while the area of non-paddy fields has increased over the past ten years.

Accuracy assessment of the derived maps

The maps that were created revealed several discrepancies between the three methods used. The error matrix has been used in a variety of land classification experiments (Bharatkar, P.S. and Patel, 2013; Comber et al., 2012; Lewis & Brown, 2001), and it was a key component of this study as well.

The derived maps were evaluated for classification accuracy using error-based matrices, and the accuracy of each land use class, as well as the omission and commission errors, overall accuracy, and kappa coefficient for each method, were calculated.

The classification of each pixel against the definite conditions of land cover, obtained from its corresponding ground truth details, is compared with the overall accuracy of the classified image (Rwanga & Ndambuki, 2017). Table 1 shows the summarized results obtained for the overall accuracy of each map derived (in each growth stage) through each method in 2008/2009 and 2018/2019.

Table 1
Summary of the overall accuracy of each method in both years

Year	Overall Accuracy%								
	Method 1			Method 2			Method 3		
	(i)	(ii)	(iii)	(i)	(ii)	(iii)	(i)	(ii)	(iii)
2008/2009	44.5	41.0	**85.5**	63.0	**84.0**	41.5	**85.5**	30.0	26.5
2018/2019	43.5	42.5	**89.5**	70.5	**88.5**	32.5	**53.0**	28.5	28.5

(i) = flooding and transplanting /initial stage, (ii) = development stage, (iii) = harvesting stage

LULC pattern of Gampaha district 2009

Figure 6 shows the land use land cover map created from Landsat 5 satellite imagery, together with a pie chart demonstrating the comparative distribution of land use categories: water bodies, paddy lands, built-up areas, forests and other vegetation.

According to the results, forests and other vegetation represented the dominant land cover type, encompassing a land area of 768.09 km², which accounted for 56% of the total study area in 2009. Paddy lands became the second most dominant land use type, covering 283.81 km^2 of land area (21% of the total area). Additionally, water bodies covered 182.91 km², accounting for 14% of the total area, while built-up areas comprised 128.38 km², or 9% of the total area. An overall accuracy of 82.9% was achieved by this classification system.

LULC pattern of Gampaha district 2019

Landsat 8 satellite images were used to create a land use and land cover map of the Gampaha district in 2019 (Fig. 7).

According to the results, forests and other vegetation represented the predominant land cover type, encompassing an area of 811.83 km², which accounts for 60% of the total study area. Built-up areas accounted for 290.60 km^2 or 21% of the total area, exhibiting the second most extensive land use type.

Water bodies (194.02 km^2, 14% of the total area) and paddy land (66.74 km^2, 5% of the total area) exhibited the minimal land cover in 2019. The overall accuracy of the classified images was determined to be 70.4%.

LULC Change Detection from 2009 to 2019

During the period from 2009 to 2019, there was a noticeable increase in the proportions of water bodies, forests, and other vegetation, as well as built-up areas, calculated at approximately 11.11 km², 43.74 km², and 162.22 km², respectively (Fig. 8). In contrast, the area identified as paddy lands experienced a drastic reduction of 217.07 km^2. The proportional change in the built-up areas was the most notable, showing a 126.4% increase, which indicates a significant alteration in the land use and land cover of the study area. Moreover, the noted decrease in paddy land area, representing a 76.5% reduction, raises substantial concerns about sustainability in the agricultural sector.

The cross-tabulation of land-use and land-cover data is presented in Table 2.

Table 2 Cross tabulation of LULC changes from 2009 to 2019

Sum of area change (km^2)	**LULC 2019**				
LULC 2009	Water bodies	Buildup areas	Forests & other vegetation	Paddy lands	**Total**
Water bodies	30.82	51.91	91.63	8.53	**182.91**
Buildup areas	17.07	60.71	46.67	3.93	**128.38**
Forests & other vegetation	104.70	106.48	523.66	33.25	**768.09**
Paddy lands	41.43	71.50	149.87	21.01	**283.81**
Total	**194.02**	**290.60**	**811.83**	**66.74**	**1363.19**

Discussion

According to Table 1, in both periods, the best-derived maps demonstrated more than 80% overall accuracy by using the "Supervised classification" method. The best map was obtained from the image obtained during the harvesting stage. This may be because during this period, the color the paddy plants distinctly varied, as they are displayed in yellow, and these results are similar to previous Gopalakrishnan et al., (2020) findings.

In both years, the "TCT and NDVI" method demonstrated more than 80% overall accuracy. According to Riadi & Suriadi, (2017), this method can be implemented to identify paddy fields that are still green. TCT functions by compressing spectral data into a few bands that correlate to physical scene characteristics (Sheng et al., 2011). The GRN component in TCT illustrates greenness, and the WET component illustrates the wetness of paddies. The combined components of GRN, WET, and NDVI led to a reliable classification of rice plants during the highest greenness stage. The maps derived through this method showed that the best map was generated during the development stage, as evidenced by the years 2008/2009 and 2018/2019, which agrees with the findings of Riadi & Suriadi (2017).

With respect to the “LSWI and NDVI” method, the maps derived in 2008/2009 demonstrated an overall accuracy of 85.5%, whereas those derived in 2018/2019 demonstrated an accuracy of 53%. The best map derived through this method was from the image downloaded during the initial stage. The LSWI is greater than the NDVI during this stage, which agrees with the Wang et al., (2015); Xiao et al., (2002, 2006) findings.

Wang et al., (2015) achieved 89.8% overall accuracy through this method conducted in Jiangsu, China, where the NDVI was greater than the LSWI during the development and harvesting stages, resulting in low overall accuracy. There may be some discrepancies in wetland misclassification as paddy land due to some similar characteristics. Zhou et al., (2016) reported flooding periods of natural wetlands and paddy fields are different. Compared with paddy wetlands, natural wetlands flood earlier and last for a relatively shorter duration. When some wetland areas that were taken from Google Earth Engine were compared with the classified paddy lands, most of them were not misclassified as paddy.

However, the results vary with the limitations of high-quality images. The discrepancy in accuracy in the derived maps during 2008/2009 and 2018/2019 using the “LSWI and NDVI method” could be attributed to differences in paddy growth stages when selecting Landsat images. The pixels with LSWI values higher than the NDVI values were classified as paddy pixels in this process. During the Maha season every year, paddy fields mostly grow in October and November, and the initial stage of paddy fields can be observed during this period. Since there were no high-quality Landsat 8 images with low cloud cover in November 2018, an image from December had to be downloaded. As a result, the NDVI values in most paddy fields are greater than the LSWI values after such a one-month delay. Using multiyear Landsat-8 data and other optical sensors, such as Landsat TM/ETM + and Sentinel-2A/B, this image data shortage could be resolved in the future (W. Wang et al., 2012)

For the "LSWI and NDVI method," the best image should be downloaded only during the months of paddy flooding and transplanting. Farmers prefer sunny days for rice transplanting because persistent cloudy/rainy days can stunt the growth of young plants, leading to dead plants (Xiao et al., 2006). Therefore, the Landsat images taken on November 4, 2008, and December 2, 2018, had better quality and less cloud cover than did the other images.

Land use and land cover (LULC) refer to the utilization of land through various activities, including agriculture, settlement development, conservation practices, urban expansion, wildlife habitats, and

other human activities (Seyam et al., 2023; W. M. D. C. Wijesinghe & Withanage, 2021). Identification of LULC dynamics is critical for assessing local, regional, and global alterations of land (Seyam et al., 2023) for the implementation of effective management strategies (Kumar & Agrawal, 2019; Seyam et al., 2023; W. M. D. C. Wijesinghe & Withanage, 2021). for utilizing natural resources, promoting sustainable approaches (Withanage et al., 2024). Therefore, researchers are involved in studying spatial and temporal LULC change factors, and patterns using highly recognized satellite images (Seyam et al., 2023; Wijesundara et al., 2023).

Changes in water bodies

The transformations in water bodies within the Gampaha district from 2009 to 2019 is shown in Fig. 9.

The analysis revealed that approximately 30.82 km² of water bodies remained unchanged, while 163.20 km² of area was newly transformed into water bodies, as indicated by the dark blue and light blue colors, respectively. Furthermore, approximately 152.09 km² of water bodies extant in 2009 were converted into non-water bodies by 2019 (Table 2), as shown in pink color. This conversion may be attributed to the filling of inland water bodies for the development of industrial areas, settlements, and infrastructure for transportation.

Despite the decline in water bodies, a considerable area has been newly designated as a water body within the study area. Sri Lanka, identified as a climate change extreme global hotspot (Somasundaram et al., 2020), encompasses numerous natural and man-made water bodies. However, the availability of surface water varies for different reasons, and extreme weather conditions may contribute to the formation of new water bodies. With the increase in impervious surfaces associated with built-up areas, the collection of waste at dumpsites may lead to the obstruction of natural drainage and block the floodplains in the study area. These areas may be misclassified as new water bodies with other class pixels during remote sensing image classification.

Nevertheless, the total water body distribution (14%) in the area has remained relatively constant throughout the observation period (Figs. 6 & 7).

Changes in the built-up areas

Figure 10 illustrates the transformations in built-up areas within the study area, specifically the Gampaha district, highlighting a significant shift in land use and land cover.

The area of built-up areas has more than doubled after 2009, increasing from 128.38 km² to approximately 290.60 km² over the decade, representing a 126.4% increase.

The dark red areas on the map designate 60.71 km² of built-up areas that remained unchanged during the period. However, approximately 229.89 km^2 of water bodies, forests, and other vegetation, and paddy lands were converted into newly established built-up areas, indicated in standard red. The pink areas

indicate the built-up areas that have transitioned into other land use categories, totaling 67.67 km² in total (Table 2).

According to the cross-tabulation of LULC data in Table 2, water bodies (51.91 km², 28% of the initial area) and paddy lands (71.50 km², 25% of the initial area) are the main categories that experienced a substantial initial land area changes, resulting in new built-up areas by 2019.

This period was accompanied by the implementation of different large-scale development projects, including the construction of expressways passing over the Gampaha district, which may have accelerated the urban expansion in the study area through new residential settlements, commercial buildings, and business places, increasing the value of the land (Sakalasooriya, 2021; W. A. M. O. Wijesinghe & Gamage, 2024)

Changes in forests and other vegetation

The changes in forests and other vegetation lands, including grasslands, crop lands, and home gardens, are illustrated in Fig. 11.

Dark green areas represent unaltered forests and other vegetation, covering approximately 523.66 km². Areas preliminarily classified as non-forests and other vegetation lands that have transformed into new forests and other vegetation areas are indicated in described by light green, totaling approximately 288.17 km^2. According to the LULC data (Table 2), 244.43 km² of initial forests and other vegetation lands were converted to other land use types, denoted by the pink color. A total of 13.9% of the initial forests and other vegetation lands decreased with new built-up areas, representing the major land use conversion, and 52.8% (149.87 km^2) of the paddy lands in 2009 were replaced by forests/ other vegetation by 2019, revealing a significant contribution to expanding the area of forests and other vegetation compared to water bodies and built-up areas.

Change detection data (Table 2) indicate that approximately 50% (91.63 km²) of the initial water bodies were converted into forests and other vegetation. This alteration may be attributed to the filling of water bodies with sediments and organic matter, as well as the gradual colonization of these areas by terrestrial plants or to misclassification with periodically flooded wetlands within the study area, where differentiating them from vegetation poses challenges in remote sensing. Overall, this class has dominated the study area, despite the vegetation of the Gampaha district being highly fragmented over the past few decades (Ranaweera et al., 2021).

Changes in paddy fields

Figure 12 illustrates the transformation in paddy fields in the Gampaha district over a decade, covering the period from 2009 to 2019.

The regions in dark green indicate that 21.01 km² of paddy land remained unchanged during this period. Light green areas represent the newly emerged paddy lands, which cover approximately 45.71 km². The

conversion of paddy land into other land classes is indicated in pink, totaling approximately 262.8 km². Specifically, 52.8% of the paddy fields that existed in 2009 were converted into forests and other vegetation, while approximately 25.1% of the original paddy fields were replaced with newly developed built-up areas.

Over time, many people have transitioned into the industrial sector, contributing to a shortage of labor in agricultural work, particularly in paddy cultivation. A lack of financial support, degradation of soil fertility, and an irregular water supply have led most farmers to abandon rice production, shifting to alternative crops such as coconuts, vegetables, and fruits. Furthermore, urbanization has led to increased population growth and human population growth, as well as elevated demand for land, following the transformation of paddy fields for commercial and residential development. These factors collectively explain the significant decline in initial paddy fields observed in the Gampaha district since 2009.

Conclusion

This study was carried out in the Gampaha district of Sri Lanka, a rapidly expanding area. In the context of remote sensing techniques for identifying paddy fields according to growth stages and detecting land use and land cover changes, this study presents a new research perspective implemented in the study area.

When considering the identification and mapping of paddy lands in the Gampaha district, all three methods used in this study can be applied, as they have achieved high accuracy (more than 80%) for each method, except for the image classified in 2018/2019 using the LSWI and NDVI methods. If there are sufficient high-quality Landsat images within the particular growth stages of a paddy plant, especially during the flooding and transplanting stage, the “LSWI and NDVI method” is the most appropriate method for identifying and mapping paddy lands. Furthermore, it can be concluded that the “TCT and NDVI method” is the most appropriate method during the development stage, and the “Supervised classification method” is the most appropriate method during the harvesting stage for identifying and mapping paddy lands in the Gampaha district.

These effective techniques can be useful for detecting paddy lands based on their growth stage in any area, for the actual detection of total lands that are preparing for cultivation, expected yield, and the number of paddy lands that are destroyed by natural or other disasters. Hence, compared with field data collection, remote sensing is a timely and cost-saving method for every season.

The study indicates that forests and other vegetation are the primary land use in the study area, Gampaha district, Sri Lanka. Between 2009 and 2019, the area covered by forests and other vegetation increased by 5.7% (43.74 km^2).

Water bodies are a secondary land use type, demonstrating a 6.1% (11.11 km²) growth over the research period; however, in terms of overall area distribution, they are the only class that remained constant (14%).

Due to the transformation into built-up areas, forests, and other vegetation, as well as the loss of water bodies, 76.5% (217.07 km²) of the initial paddy lands have decreased, indicating a significant reduction in the study area. A significant conversion was observed in the buildup areas of the study area, increasing by 126.4% (162.22 km²) due to the expansion of human settlements, industries, and new expressway development over the ten-year period from 2009 to 2019.

The study's findings suggest that stakeholders and policymakers in the region and country should be aware of the changing patterns of LULC and the relationships between potential and current issues in various social, environmental, and economic sectors of the country, as well as the consequences of different development strategies.

However, in this study, 30 m resolution, medium-quality free Landsat satellite images were used, which may be suitable for small-scale studies but not for detailed assessments. Therefore, if possible, the use of high-resolution images for future land investigations is recommended for improved accuracy.

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Figures



Figure 1

Study area - Gampaha district, Sri Lanka

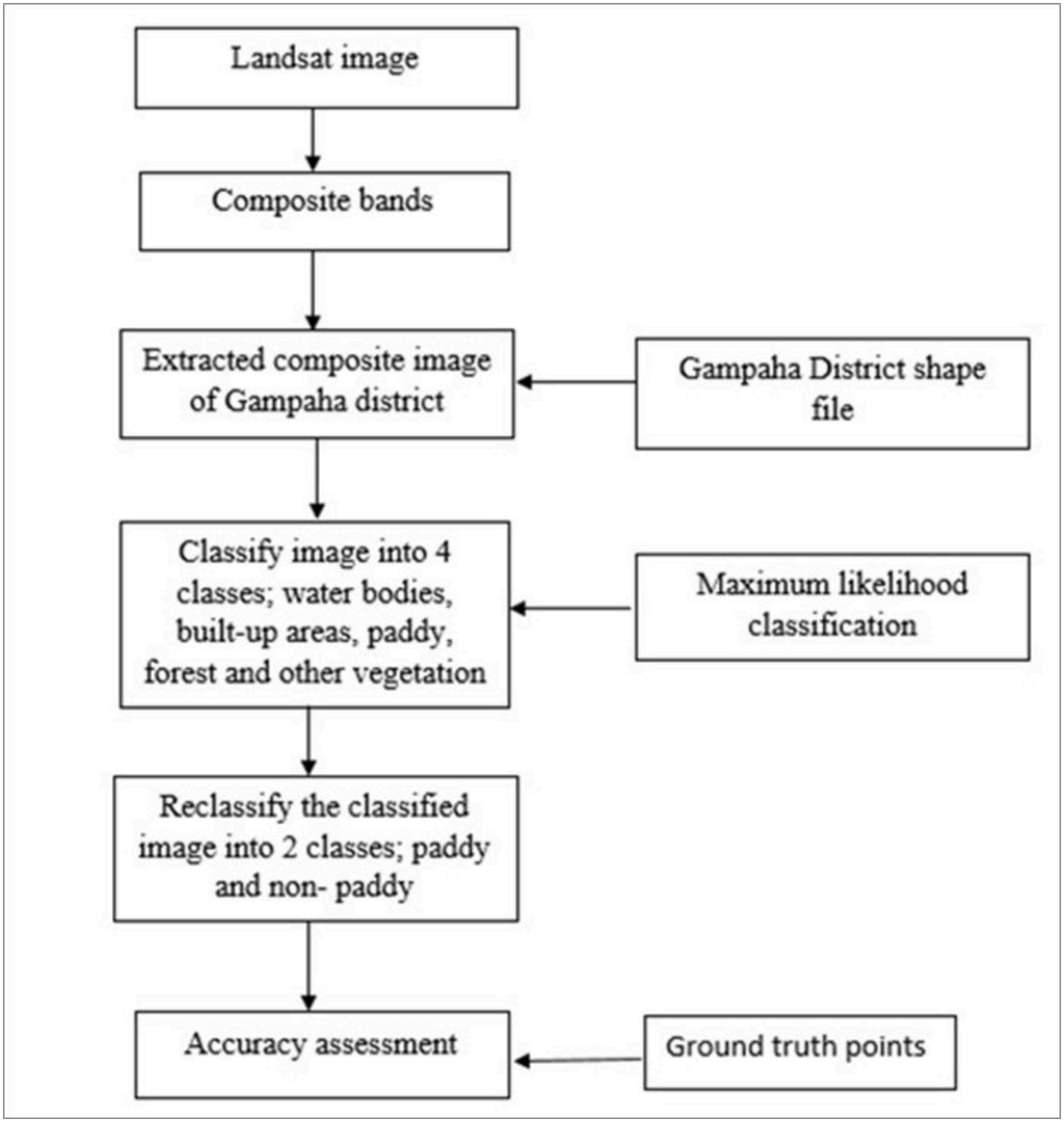


Figure 2

Flowchart of the Supervised classification method

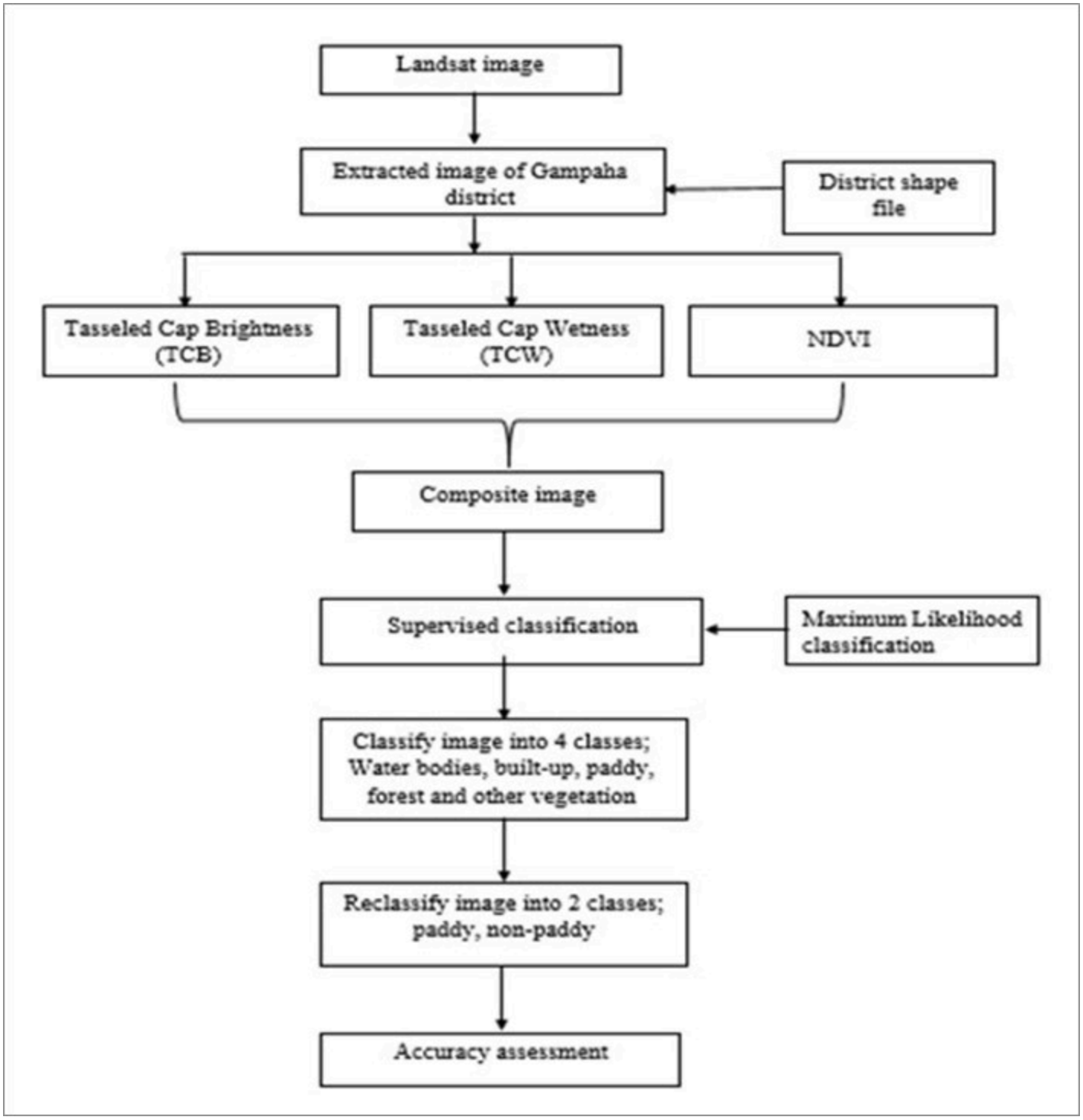


Figure 3

Flowchart of TCT and NDVI method

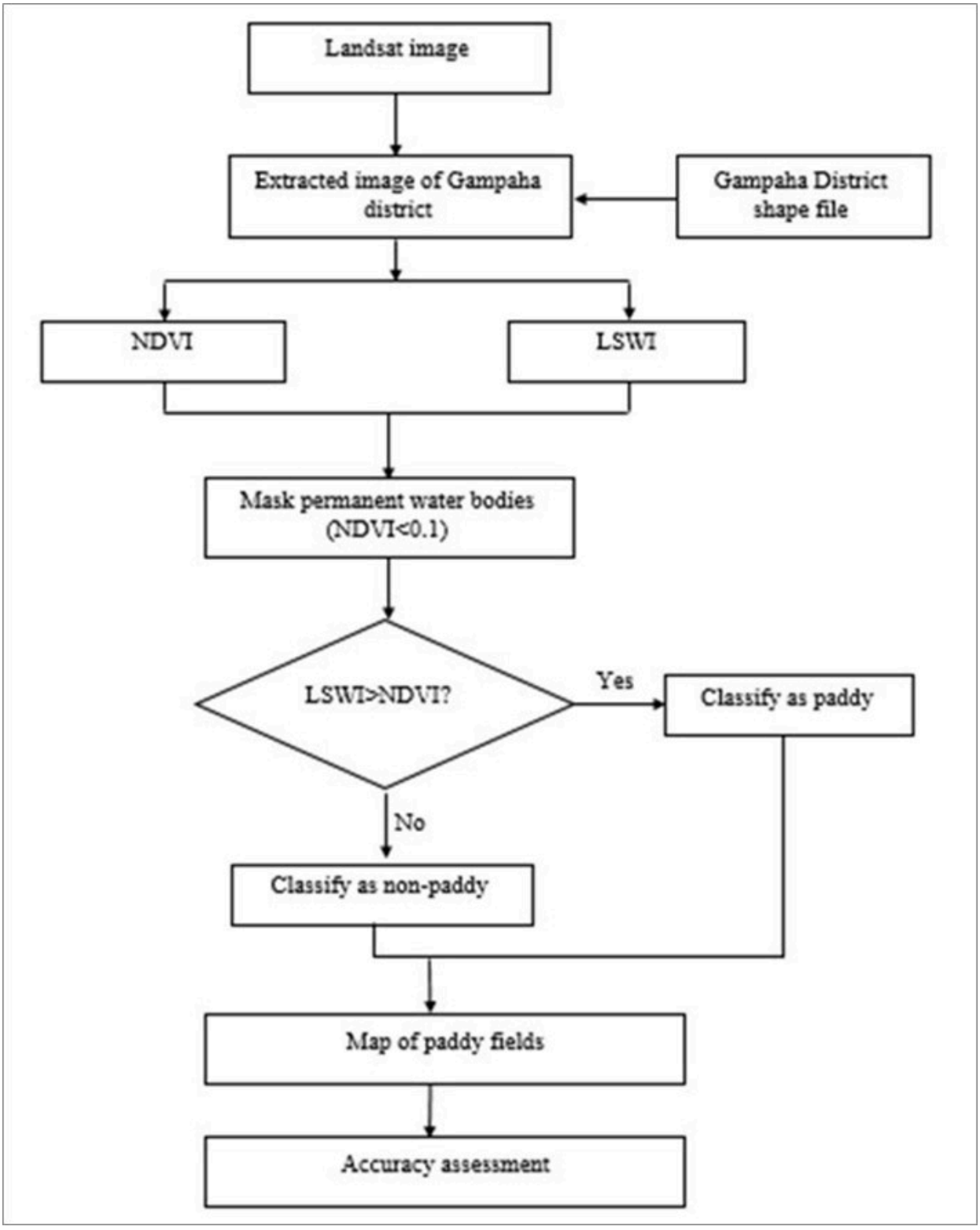


Figure 4

Flowchart of the LSWI and NDVI method

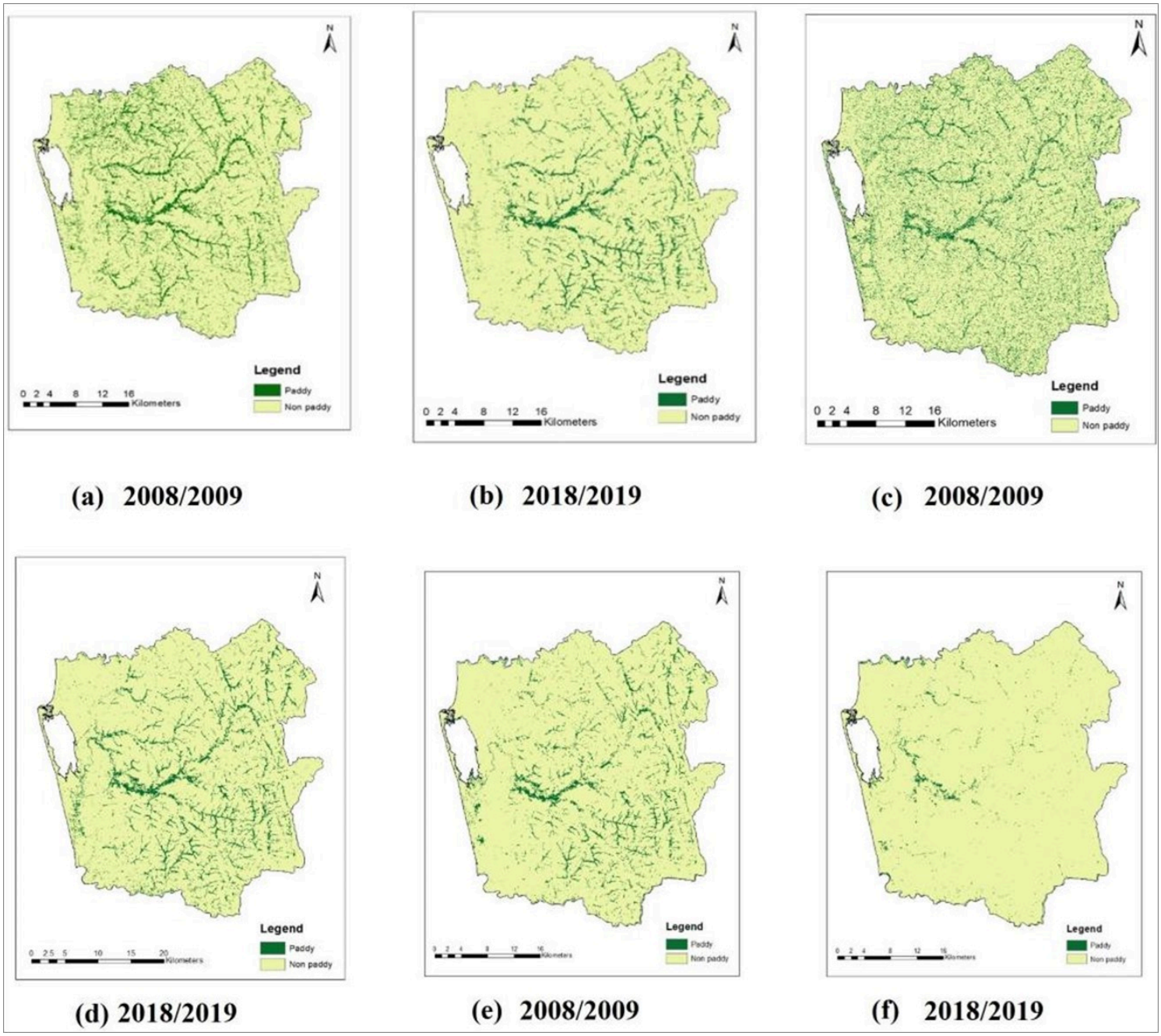
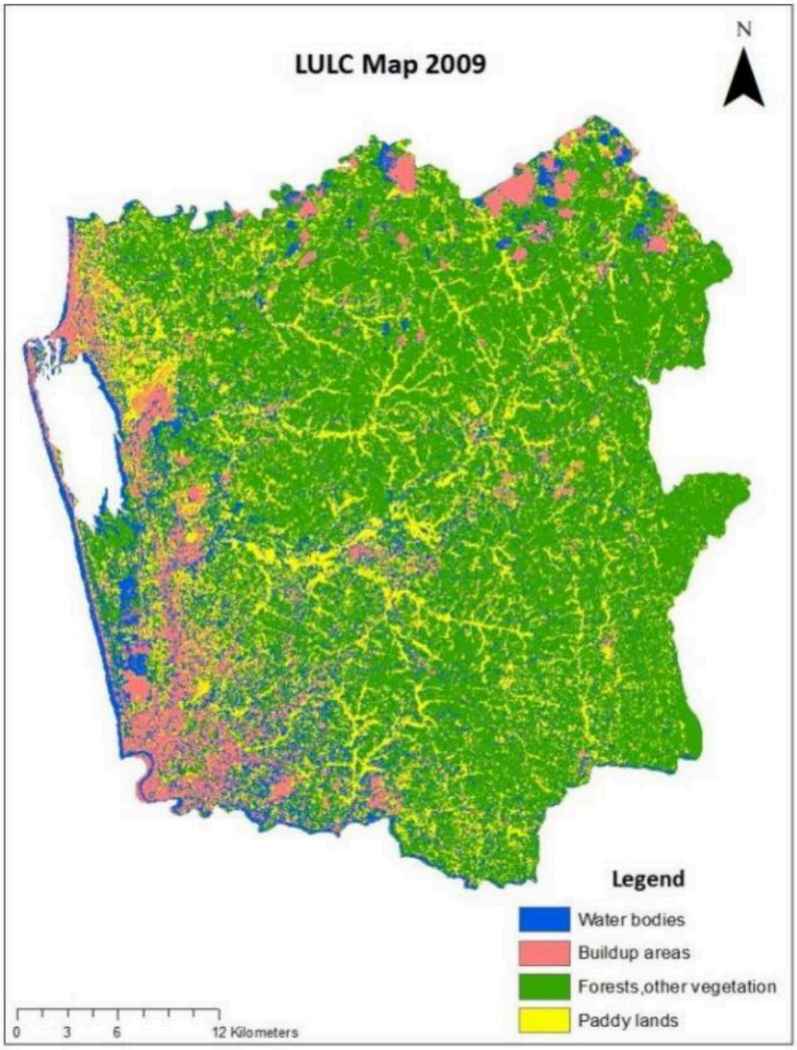


Figure 5

Derived maps through the (a, b) supervised classification, (c, d) TCT and NDVI, and (e, f) LSWI and NDVI methods



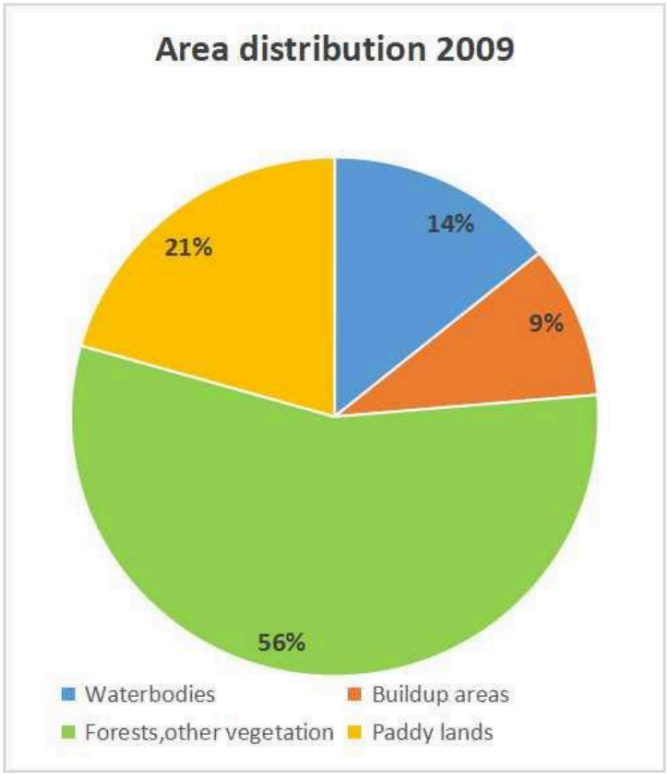


Figure 6

LULC Map and area distribution of the Gampaha district in 2009

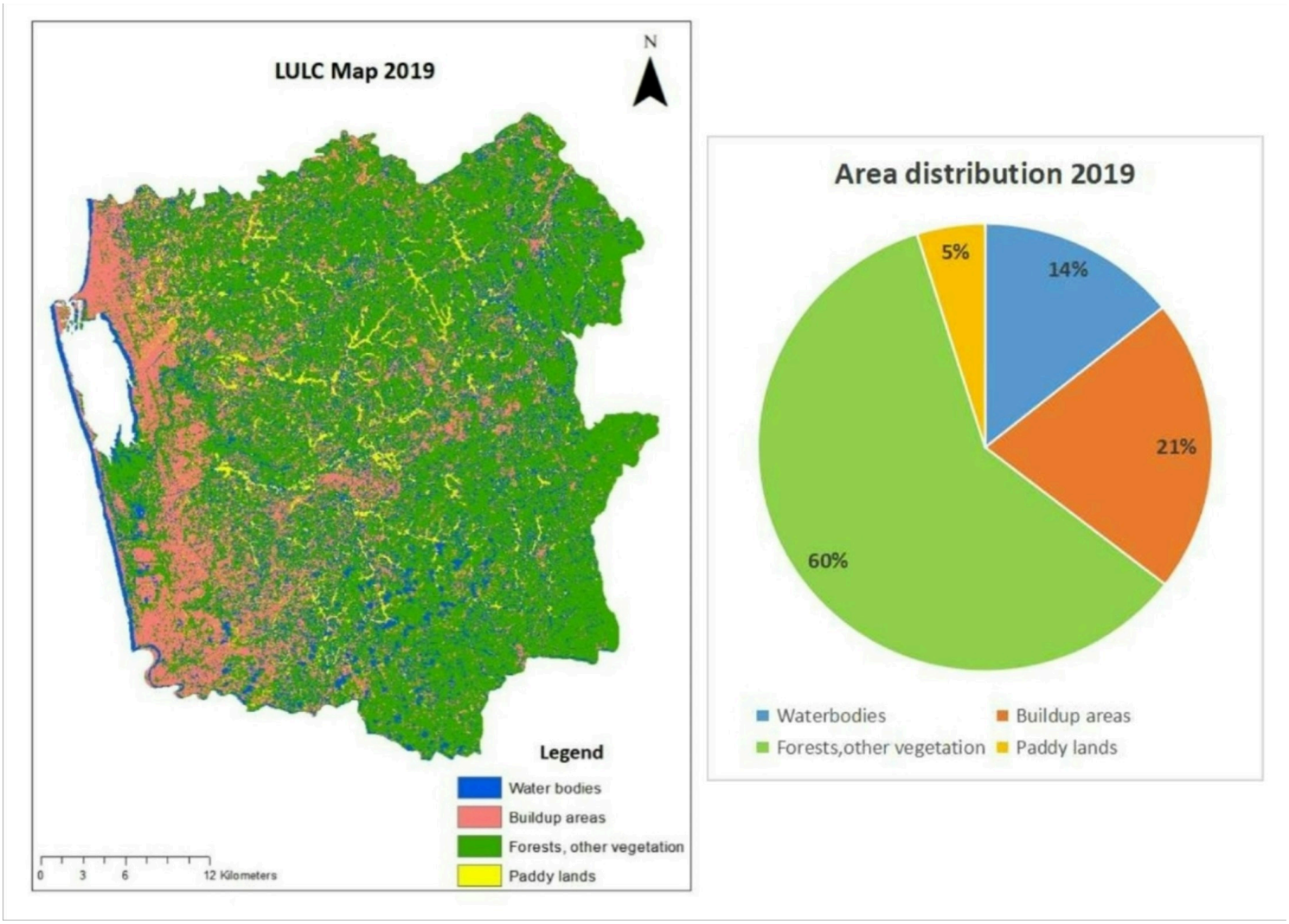


Figure 7

LULC Map and area distribution of the Gampaha district in 2019

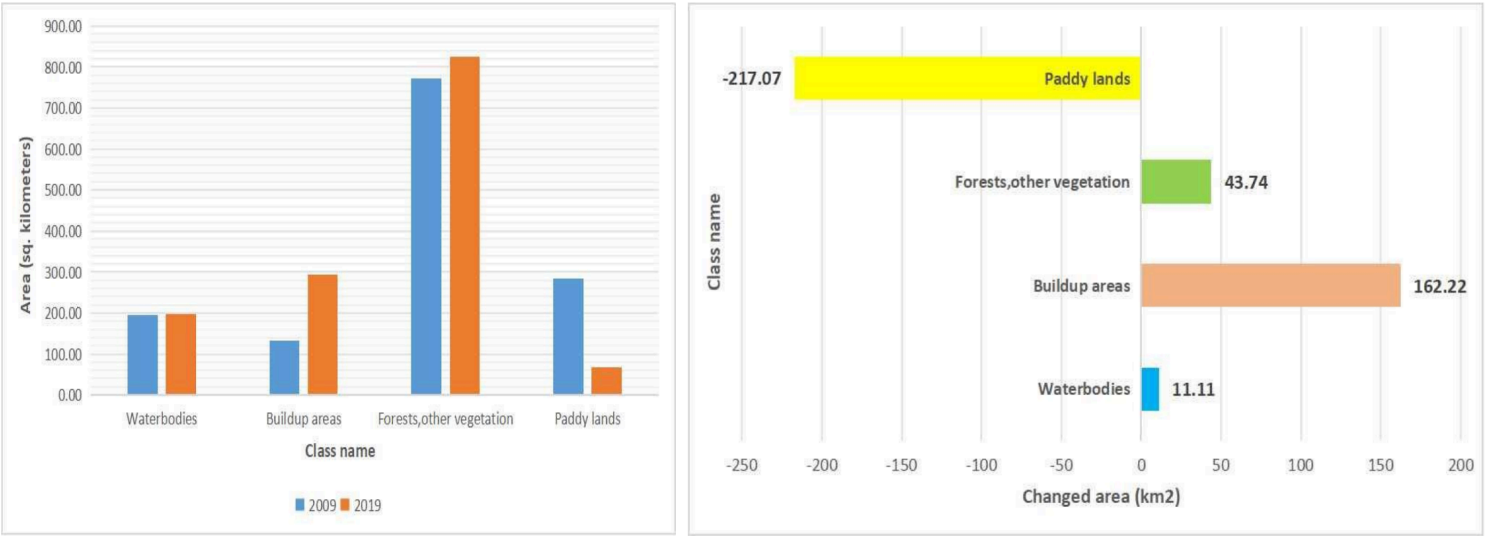


Figure 8

LULC comparison and area change graph in 2009 and 2019

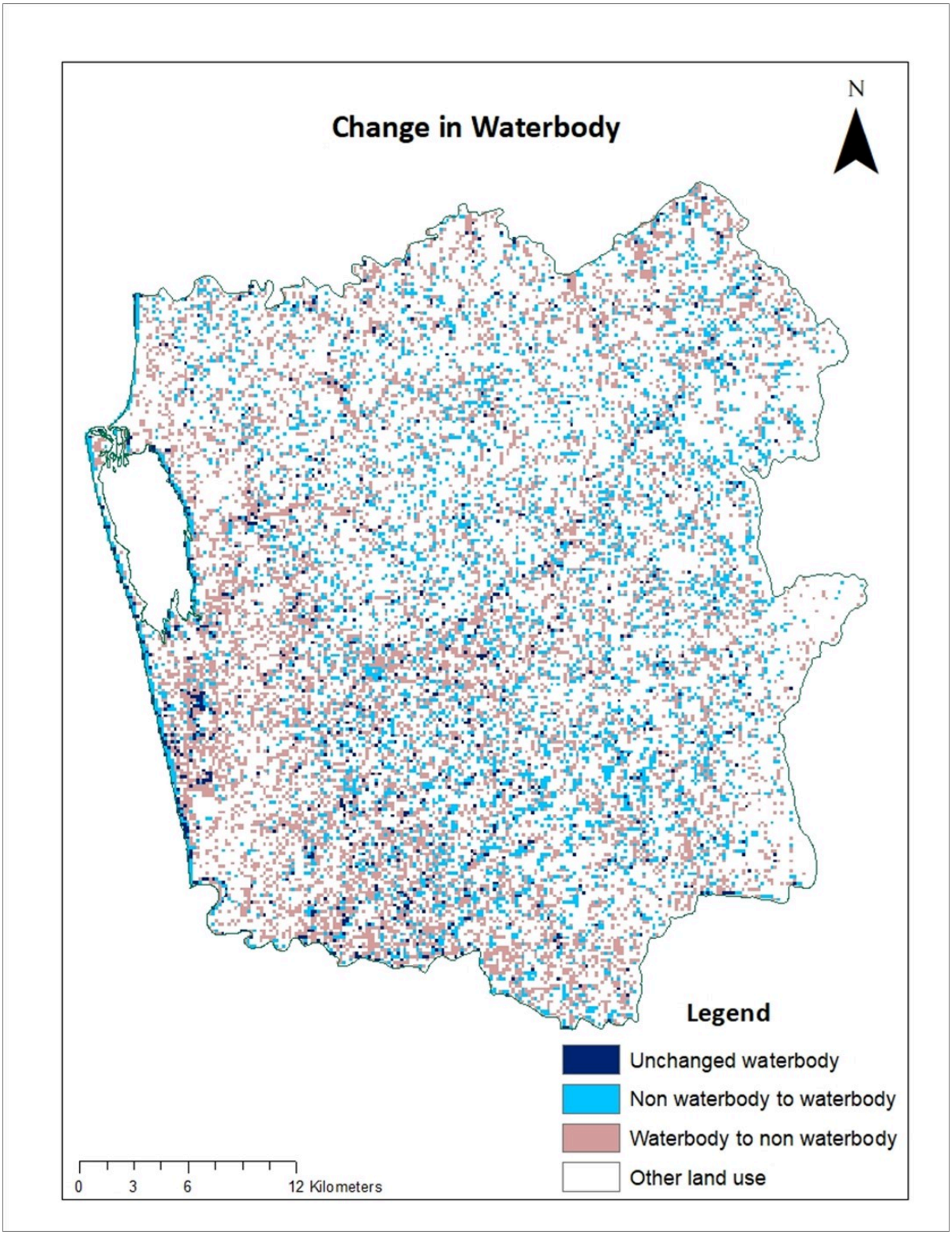


Figure 9

Changes in water bodies from 2009 to 2019

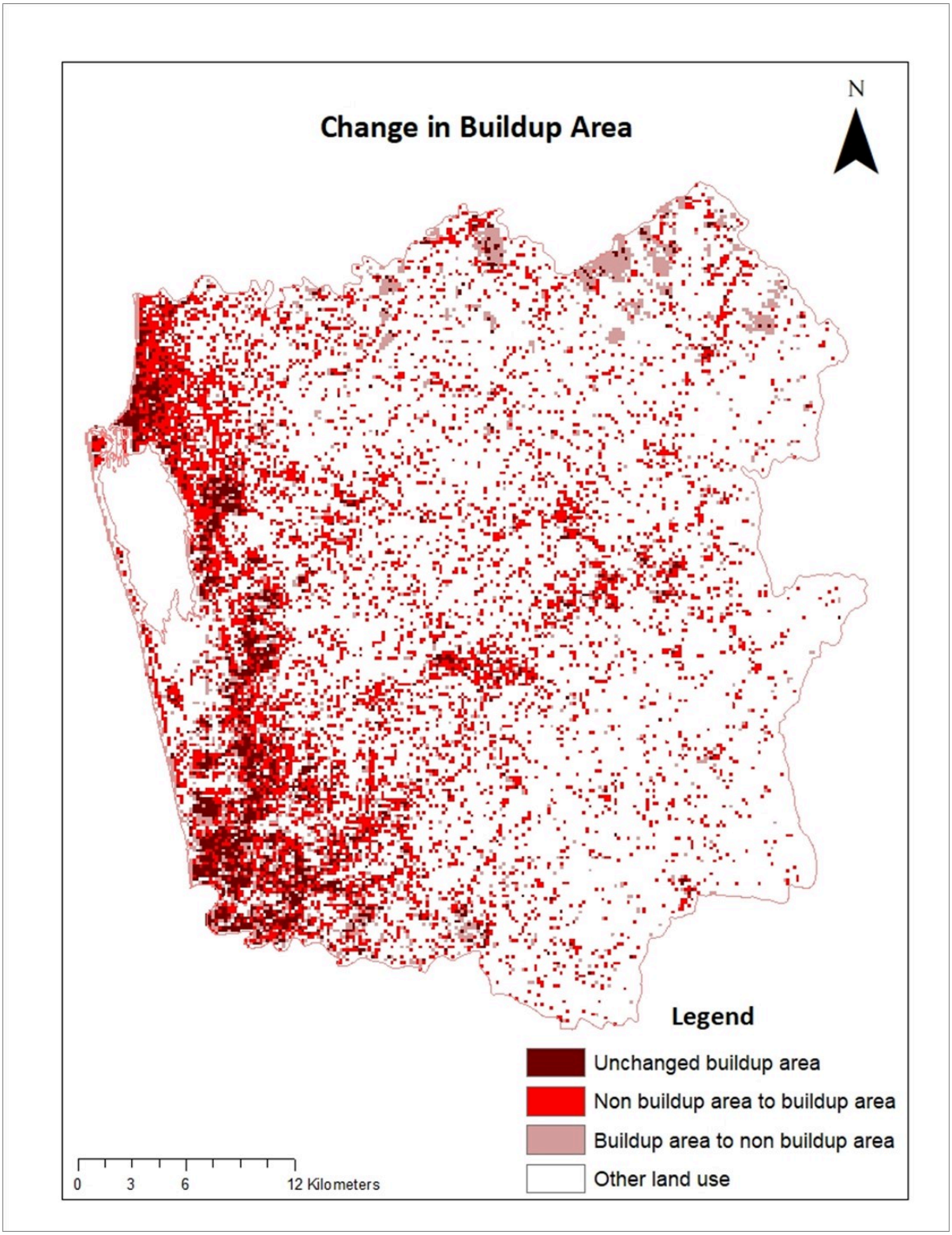


Figure 10

Changes in built-up areas from 2009 to 2019

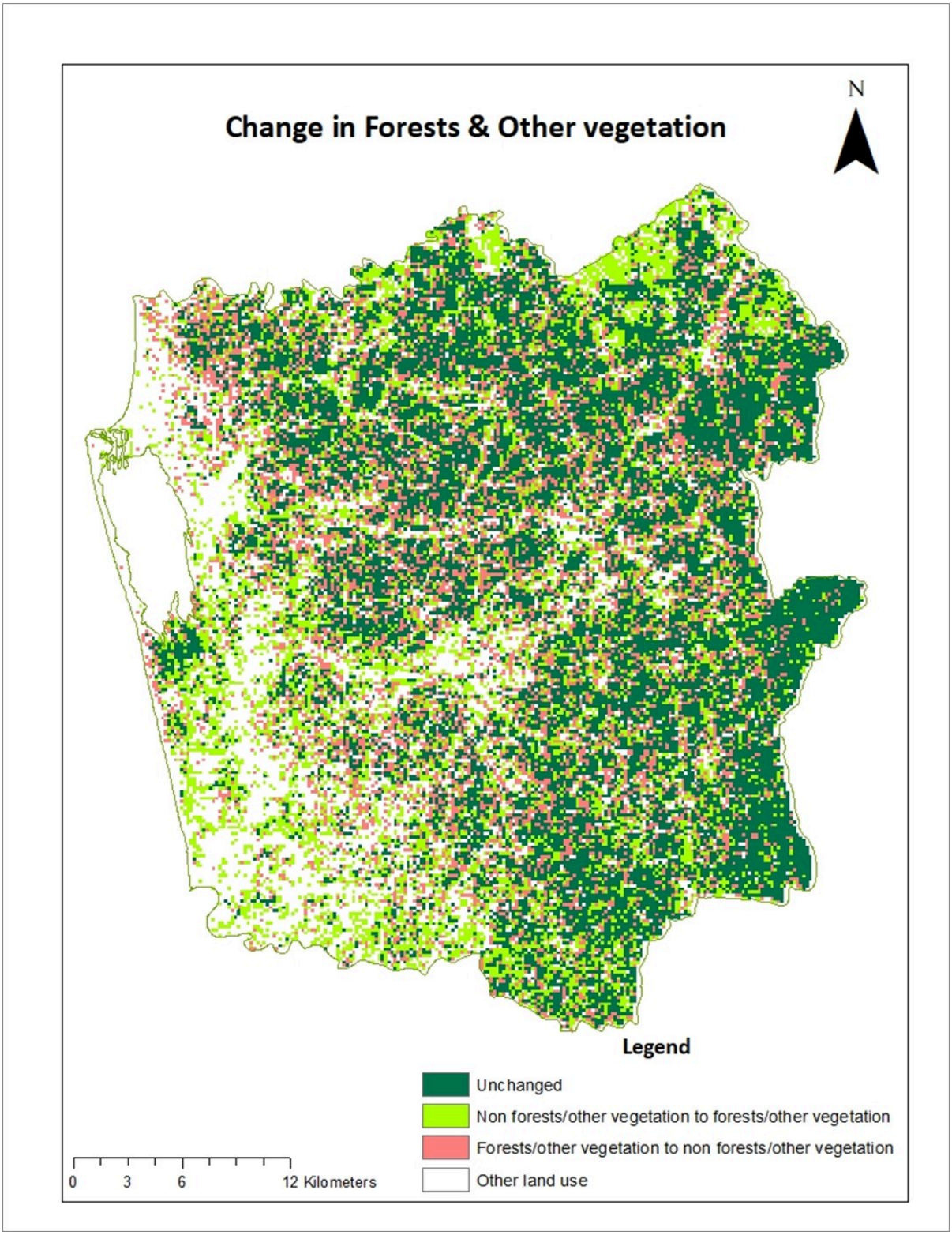


Figure 11

Changes in forests and other vegetation types from 2009 to 2019

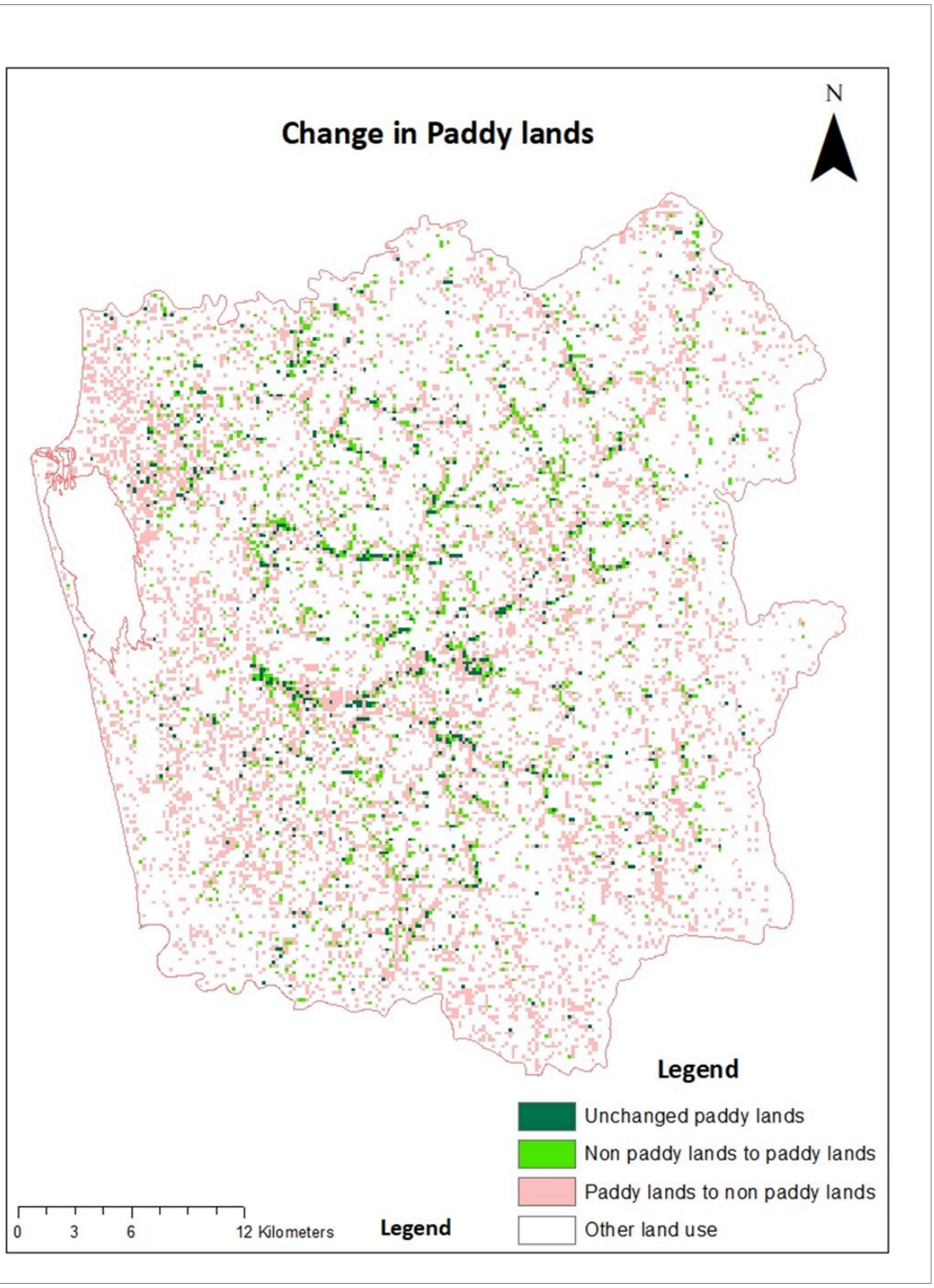


Figure 12

Changes in paddy fields from 2009 to 2019